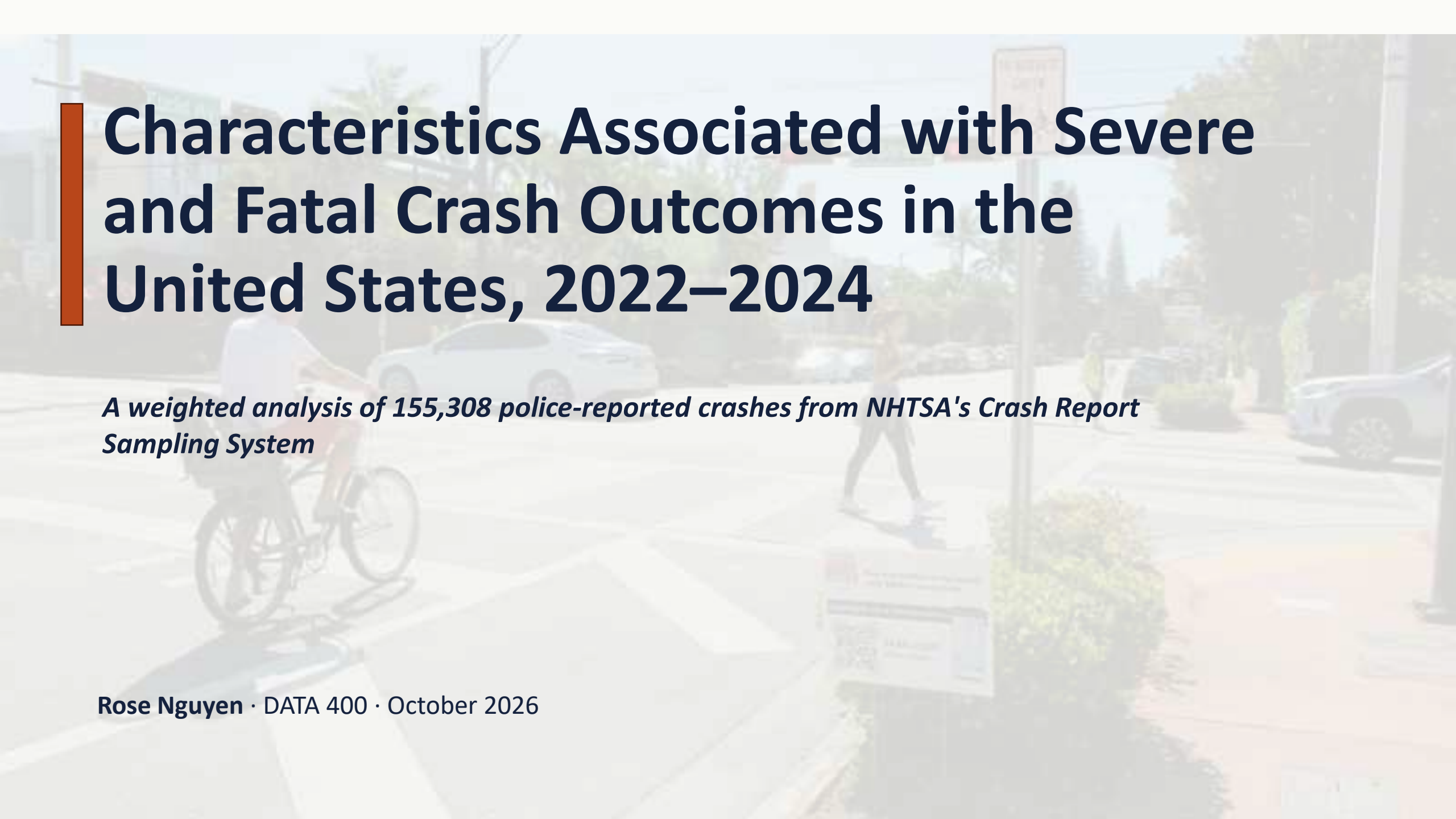


The background of the slide is a faded, light-colored photograph of a city street. In the foreground, a person is riding a bicycle on the left side of the frame. To their right, a pedestrian is walking across a crosswalk. In the background, there are parked cars, trees, and a street sign. The overall scene is a typical urban environment.

Characteristics Associated with Severe and Fatal Crash Outcomes in the United States, 2022–2024

A weighted analysis of 155,308 police-reported crashes from NHTSA's Crash Report Sampling System

RESEARCH QUESTION

Which behavioral and environmental characteristics are most strongly associated with severe or fatal outcomes in U.S. police-reported crashes, 2022–2024?

Behavioral

Speeding
Alcohol involvement

Time

Night (21:00 - 05:59)
Weekend
Hour and day of week

Environment

Weather
Lighting

Roadway

Urban vs rural
Junction type
Interstate

DISCIPLINARY MOTIVATION

This project comes from my focus on transportation, traffic and safety analysis. In this field, the main measure of roadway safety is not how many crashes happen but how many people are killed or seriously injured.

DATA

Three years of NHTSA's Crash Report Sampling System (2022-2024)

155,308

crashes analyzed

6.1 M

U.S. crashes per year
they represent

3.2%

are severe or fatal
(weighted)

148,966

crashes in the regression

ACCIDENT file · one row per crash

Severity, alcohol, time, lighting,
weather, urban/rural, junction,
interstate, sampling weight

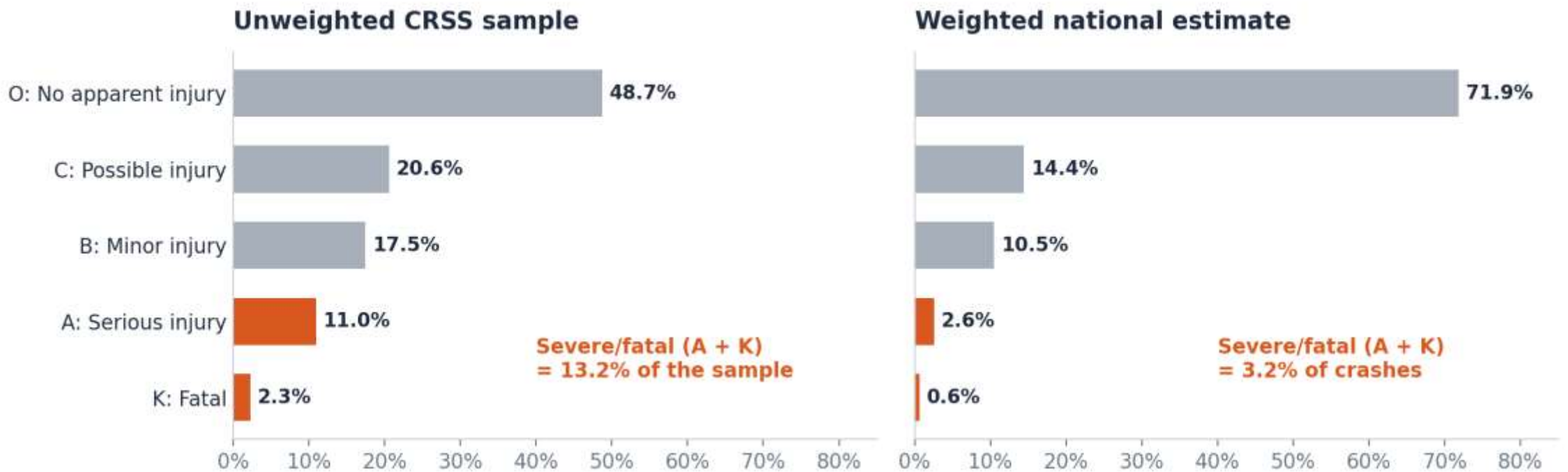
VEHICLE file · one row per vehicle

Speeding status, aggregated to one value
per crash

WHY THE SAMPLING WEIGHTS MATTER

13% of sampled crashes are severe, but only 3.2% nationally

Crash outcomes by maximum injury severity, 2022-2024



DATA CLEANING

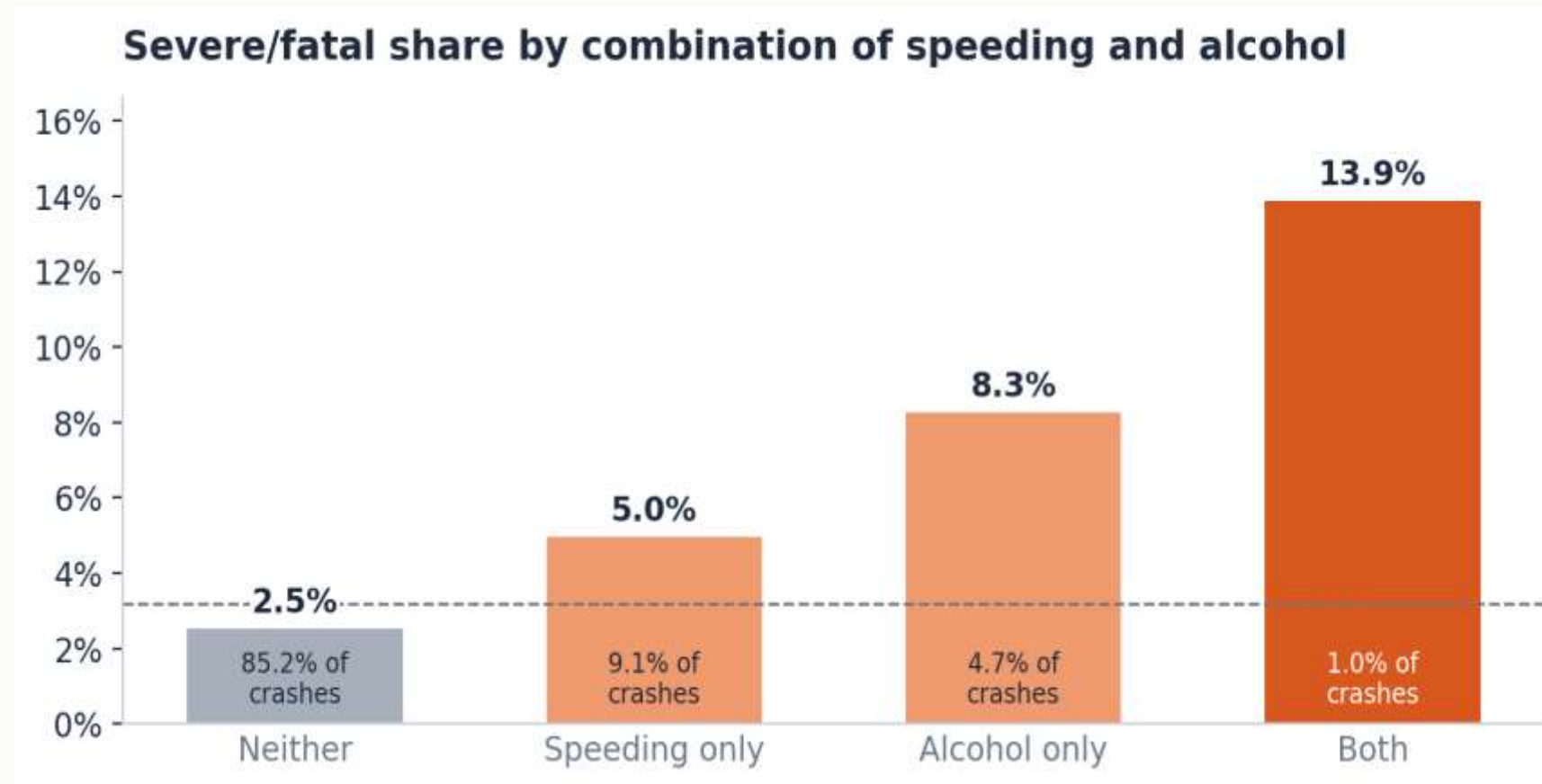
From two raw files to one crash-level dataset



Speeding rule: a crash is "Yes" if any vehicle was speeding, "Unknown" if none was but at least one vehicle's status was unknown, otherwise "No". Alcohol uses NHTSA's imputed variable, because police leave it unknown in about 39% of reports.

EXPLORATORY ANALYSIS • BEHAVIOR

Speeding and alcohol compound: 13.9% of crashes with both are severe



3.3×

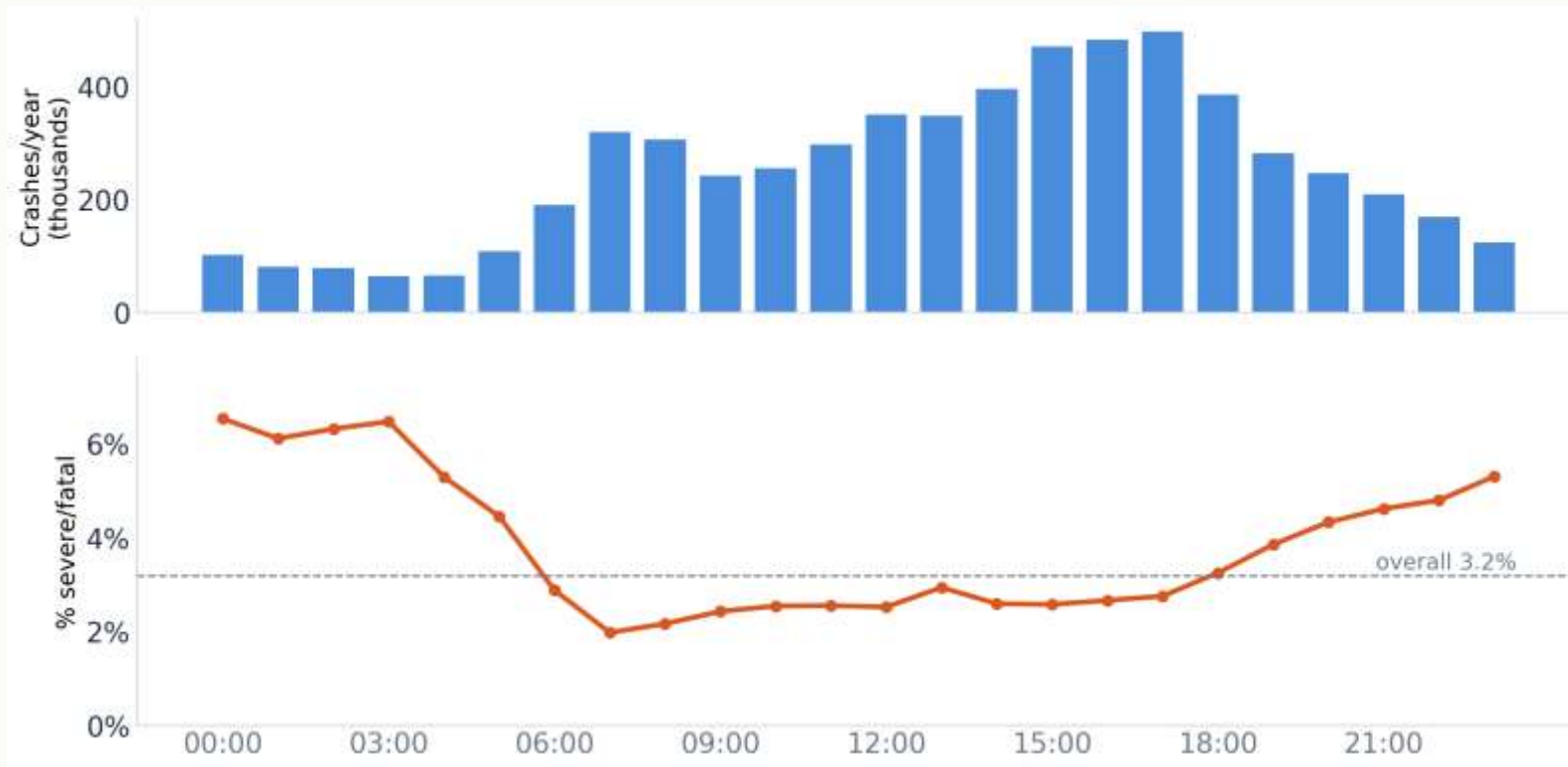
higher severe share with
alcohol (9.4% vs 2.8%)

2.1×

with speeding (5.9% vs 2.8%)

EXPLORATORY ANALYSIS · TIME

Crash volume peaks in the afternoon, while the severe/fatal share peaks after midnight



Volume

Most crashes happen in the afternoon rush, 15:00-18:00

Severity

6.5% severe at midnight vs 2.0% at 07:00

Night & weekend

5.3% vs 2.8% at night

4.2% vs 2.9% on weekends

EXPLORATORY ANALYSIS · ENVIRONMENT

Bad weather means less severe crashes, while dark, rural roads mean more

Weather · % of crashes that are severe/fatal

3.3%

Clear weather

2.5%

Rain (lower)

1.9%

Snow or ice (lowest)

Lighting and roadway · % of crashes that are severe/fatal

5.6%

Dark, unlighted (daylight 2.6%)

4.1%

Rural (urban 2.9%)

3.9%

Non-junction (intersection 2.8%)

MODEL

A weighted logistic regression, built in three steps

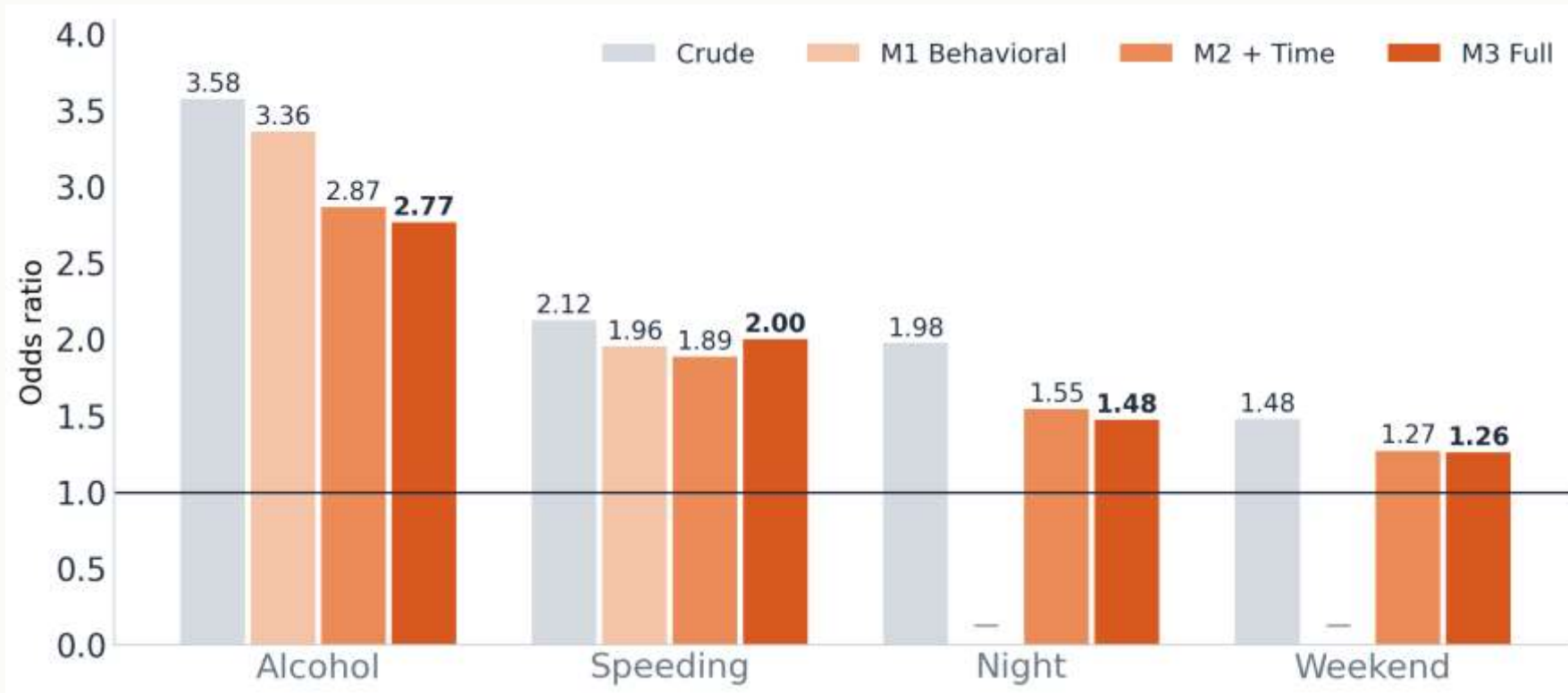
Model	Predictors (reference level)
Crude	Each predictor on its own
M1 · Behavioral	Speeding (no) · Alcohol (no)
M2 · Add Time	M1 + Night (day/evening) · Weekend (weekday)
M3 · Full	M2 + Weather (clear) · Rural (urban) · Junction (non-junction) · Interstate (no) · Year (2022)

How it was estimated

- Outcome: severe/fatal (K or A) vs. other crashes.
- CRSS weights, normalized to a mean of 1 Standard errors clustered by sampling area (67 PSUs).
- Checked against full design (strata + PSUs): main results hold.
- Night, not lighting: 95% of night crashes are in the dark.

RESULTS · MODEL BUILDING

Alcohol and speeding hold up, night and weekend shrink

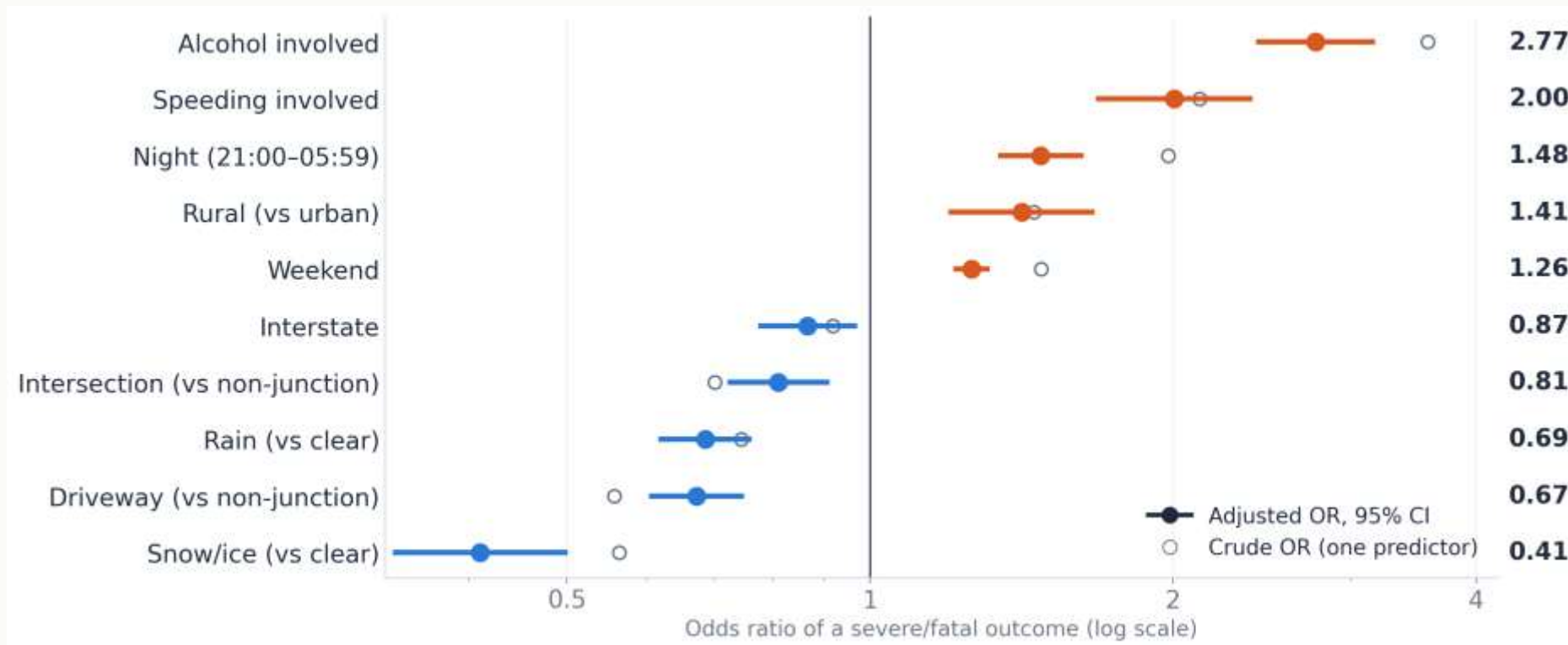


Night 1.98 → 1.48:
partly who drives at
night and how

Weekend 1.48 → 1.26:
partly because
weekend crashes
happen at night

RESULTS · FINAL MODEL

Alcohol and speeding carry the highest adjusted odds



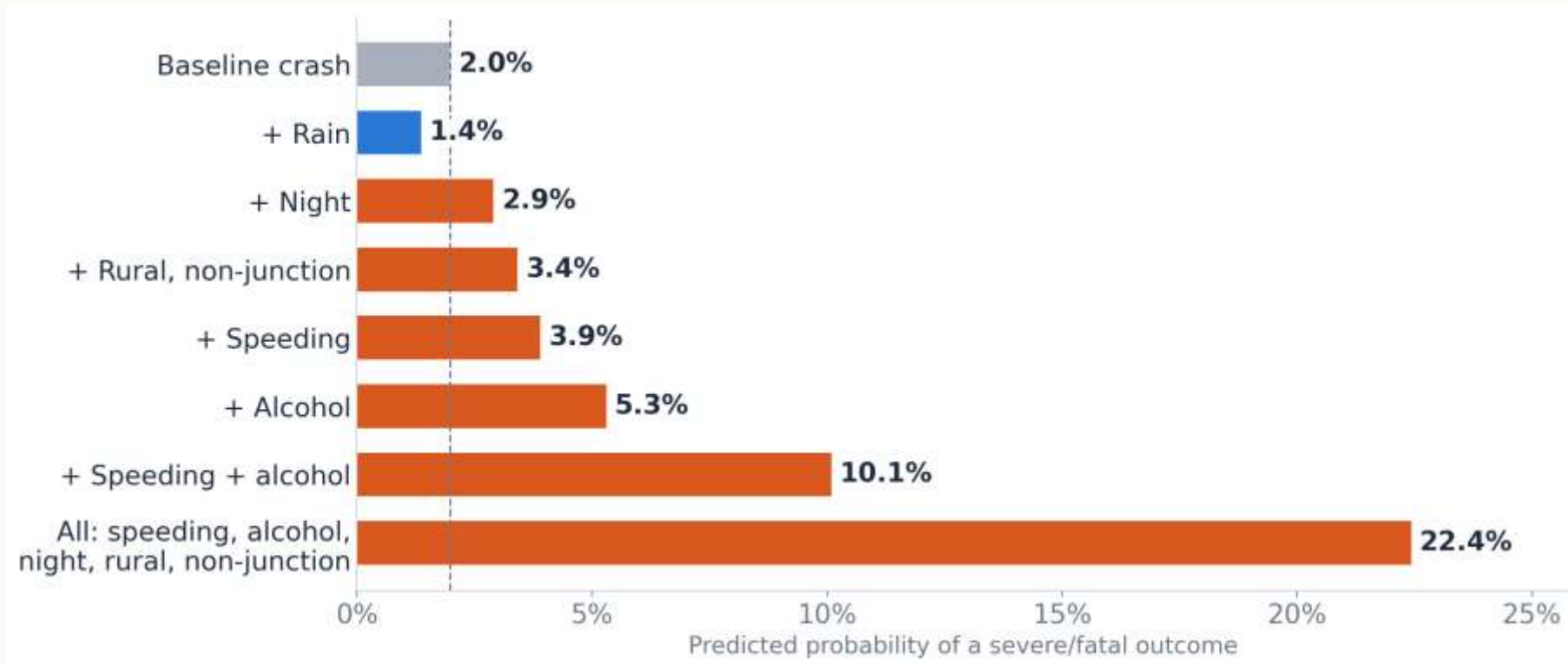
OR 2.00 = twice the odds of a severe outcome, all else equal

Hollow = crude OR

All shown ORs significant at $p < 0.05$ · PSU-clustered 95% CI

RESULTS · PREDICTED PROBABILITIES

Combining risk factors takes a crash from 2% to 22%



Baseline: 2024
weekday, daytime,
clear, urban
intersection, no
speeding or alcohol

FINDINGS SUMMARY

What is associated with a crash being severe or fatal

Behavior matters most:

Alcohol and speeding show the largest differences, and crashes with both are the most severe of all.

Night and darkness matter:

Severity peaks after midnight, especially on weekends, and on unlighted roads.

Bad weather is not more severe:

Rain and snow/ice crashes are less severe, likely because drivers slow down.

Roadway:

Rural crashes are more severe, junction crashes are less severe than open road.

STAKEHOLDERS AND IMPLICATIONS

Who is the audience for this project

**National agencies: NHTSA and the U.S.
Department of Transportation**

**State departments of transportation and state
highway safety offices**

**Local governments and metropolitan planning
organizations (MPOs)**

The traveling public

LIMITATIONS AND NEXT STEPS

What this analysis can and cannot say

Limitations

- Observational: associations, not causes
- Regression p-values are approximate
- Police-reported crashes only
- 4% with unknown status left out of the model
- No driver, vehicle or speed-limit variables

Next steps

- Add the PERSON file: driver age, seat-belt use
- Add posted speed limit and vehicle type
- Test interactions (speeding × rural)
- Full survey-design model for all estimates

| Thank You!

